10-1 Types of Data and Analysis

Point!

1 Types of Data

(1) ([1]**Quantitative data**) : Data that is expressed in numbers and where the size of the numbers has meaning.

[1] ([2]**Interval scale**) : Data where the intervals between numbers have meaning.
<Examples> Time, temperature, calendar year, etc.

[2] ([3]**Ratio scale**) : Data where not only the intervals between numbers but also the ratios of the numbers have meaning.
<Examples> Length, weight, age, etc.

(3) ([4]**Qualitative data**) : Data used to distinguish classifications or types.

[1] ([5]**Nominal scale**) : Data that only holds meaning in distinguishing different categories.
<Examples> Blood type, prefecture, phone number, etc.

[2] ([6]**Ordinal scale**) : Data in which the order is meaningful but the intervals are not meaningful.
<Examples> Grades on report cards, test rankings, grade level, etc.

3 Data Analysis Approachs

(1) ([7]**Discriptive Statistics**) : Summarizing and organizing historical data to get insights about the characteristics of a sample

[1] ([8]**Measures of centeral tendency**) : Mean, Median,and Mode.

[2] ([9]**Measures of Spread**) : Range, Variance,and Standard Deviation

(2) ([10]**Inferential Statistics**) : Used to make inferences, predictions, and generalizations about the entire population from the data of a sample.

[1] ([11]**Hypothesis Testing**)

[2] ([12]**Confidence Intervals**)

Warm Up

Answer the following questions.

(1) For the data shown in [1] to [6] below, answer A for quantitative data and B for qualitative data.

[1] Time [2] Blood type [3] Names of U.S. states
[4] Weight [5] Test ranking [6] Age

(2) Into which scale of measurement can the data in [1] to [6] be classified? Choose one from the options **A** to **D** below, and answer using the letters.

[1] Time [2] Blood type [3] Names of U.S. states
[4] Weight [5] Test ranking [6] Age

A Interval scale **B** Ratio scale **C** Nominal scale **D** Ordinal scale

Explanation

(1) [1] A [2] B [3] B [4] A [5] B [6] A
(2) [1] **A** [2] **C** [3] **C** [4] **B** [5] **D** [6] **B**

Try

1 Answer the following questions.

(1) For the data shown in [1] to [9] below, answer A for quantitative data and B for qualitative data.

[1] Marathon times [2] License plate numbers [3] Number of cars
[4] Sales revenue of a company [5] Impressions of cooking [6] Days of the week
[7] Length of traffic jams [8] Addresses [9] Favorite foods

(2) Choose the description for the characteristics of scales of measurements [1] to [4] from the options **A** to **D** in [Group A]. Then, choose the item that represents the scale from the options a to d in [Group B]. Answer both questions using the letters.

[1] Interval scale [2] Ordinal scale [3] Ratio scale [4] Nominal scale

[Group A]

A A scale with equal intervals but without meaningful ratios
B A scale where the order has meaning, but the intervals do not
C A scale represented by names or characteristics for classification and differentiation
D A scale where the origin is defined and both intervals and ratios have meaning

[Group B]

a Grade b Product prices c Phone numbers d Western calendar years

Exercise

1 Answer the following questions.

(1) Choose the correct explanation for nominal data from the options **A** to **D** below, and answer using the letter.

A Data that is expressed in numbers and where the size of the numbers has meaning.

B Data published on the internet by governments or businesses that is freely accessible and usable by businesses.

C Data to which numbers are assigned that have a meaningful order, but where the intervals have no meaning.

D Data represented by names or characteristics for distinguishing and classifying subjects.

2 Answer the following questions.

(1) Choose one scale of measurement for each of the data types shown in [1] to [6] below from the options **A** to **D** below, and answer using the letters.

[1] Temperature

[2] Occupation

[3] Number of visitors to an amusement park

[4] Uniform numbers of baseball players

[5] Number of cars sold

[6] Marathon ranking

A Interval scale **B** Ratio scale **C** Nominal scale **D** Ordinal scale

10-2 Data Analysis [1]

Point!

Data Organization

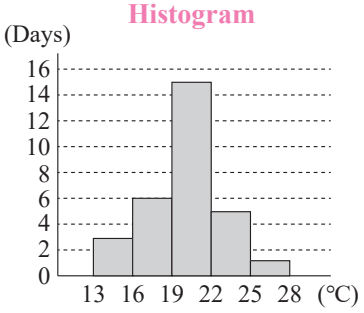
(1) ([1]**Frequency distribution table**) : A table that organizes and summarizes data, as shown on the right.

- The interval of the frequency distribution table is referred to as a ([2]**class**), the width of the interval is called the ([3]**class width**), and the midpoint value of the class is called the ([4]**class mark**).
- The number of data values included in each class is referred to as the ([5]**frequency**).

Frequency distribution table

Class (°C)	Frequency (days)
13 or higher and lower than 16	3
16 to 19	6
19 to 22	15
22 to 25	5
25 to 28	1
Total	30

Histogram



(2) ([6]**Histogram**) : A frequency distribution table presented as a bar graph. (see figure above)

(3) Measure of central tendency

Measure of central tendency : A single numerical value representing the characteristic of a dataset.

[1] ([7]**Mean**) : The value obtained by dividing the sum of the data by the number of data points. In spreadsheet software, it can be determined using the ([8]**AVERAGE**) function.

[Cell I3]=AVERAGE(B3:H3)

	A	B	C	D	E	F	G	H	I
1	Analysis of maximum temperatures in the 4th week of February 2023								
2		19th (Sun.)	20th (Mon.)	21st (Tue.)	22nd (Wed.)	23rd (Thurs.)	24th (Fri.)	25th (Sat.)	Mean
3	Tokyo	18.5	14.7	9.2	10.4	10.4	12.1	12.7	12.57
4	Osaka	14.3	12.7	8.5	10	12.5	8.5	10	10.93

[2] ([9]**Median**) : The value that appears at the center when all the data is arranged in ascending order. If the number of data points is even, the median is the mean of the two central values. In spreadsheet software, it can be determined using the ([10]**MEDIAN**) function.

[Cell I3]=MEDIAN(B3:H3)

	A	B	C	D	E	F	G	H	I
1	Analysis of maximum temperatures in the 4th week of February 2023								
2		19th (Sun.)	20th (Mon.)	21st (Tue.)	22nd (Wed.)	23rd (Thurs.)	24th (Fri.)	25th (Sat.)	Medium
3	Tokyo	18.5	14.7	9.2	10.4	10.4	12.1	12.7	12.10
4	Osaka	14.3	12.7	8.5	10	12.5	8.5	10	10.00

[3] ([11]**Mode**) : The value that appears most frequently. When arranging data in a frequency distribution table, the class mark of the class with the highest frequency is considered the mode. In spreadsheet software, it can be determined using the ([12]**MODE**) function.

[Cell I3]=MODE(B3:H3)

	A	B	C	D	E	F	G	H	I
1	Analysis of maximum temperatures in the 4th week of February 2023								
2		19th (Sun.)	20th (Mon.)	21st (Tue.)	22nd (Wed.)	23rd (Thurs.)	24th (Fri.)	25th (Sat.)	Mode
3	Tokyo	18.5	14.7	9.2	10.4	10.4	12.1	12.7	10.40
4	Osaka	14.3	12.7	8.5	10	12.5	8.5	10	8.50

Warm Up

Answer the following questions.

(1) The data on the right represents the results of a handball throw from a class of 32 students. Answer [1] and [2].

[1] Create a frequency distribution table where the class interval is 12 m to less than 15 m, with each class having a width of 3 m.

[2] Based on the frequency distribution table created in question [1], draw a histogram.

15	20	21	23	30	27	27	16
20	18	22	19	20	21	12	21
24	27	22	26	25	32	24	21
21	23	20	16	23	21	21	24

The unit is m.

(2) The data on the right represents the scores of 12 students who played a game. Determine the mean, median, and mode of the data.

6	9	3	6	10	12
4	3	8	10	7	6

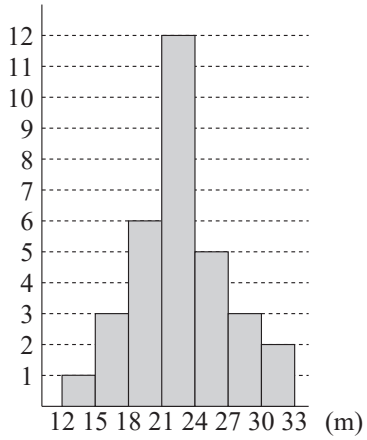
The unit is points.

Explanation

(1) [1]

Class (m)	Frequency (persons)
12 or higher and lower than 15	1
15 to 18	3
18 to 21	6
21 to 24	12
24 to 27	5
27 to 30	3
30 to 33	2
Total	32

[2] (Persons)



(2) Mean: $= \dfrac{6+9+3+6+10+12+4+3+8+10+7+6}{12} = \dfrac{84}{12} = 7$ 7 points

Median: First, arrange all the data in ascending order.

3, 3, 4, 6, 6, 6, 7, 8, 9, 10, 10, 12

Since the number of data points is even, the median is the mean of the two central values, which are 6 and 7.

Therefore, $\dfrac{6+7}{2} = 6.5$ 6.5 points

Mode: The value that occurs most frequently is 6 6 points

Try

1 The following data represents the quiz scores of 20 high school students. Answer the following questions.

3	4	9	7	6	10	5	5	7	9	
6	8	1	5	7	10	8	6	3	7	(Points)

(1) Determine the mean of the data.
(2) Determine the median of the data.
(3) Determine the mode of the data.

2 The following data shows the recorded daily maximum temperatures in Tokyo for February 2023. Answer the following questions.

13.1	9.2	6.2	11.2	13.6	15.4	11.7	10.6	3.5	12
14.1	16.9	10.3	10.7	7.8	9.6	10.8	15	18.5	14.7
9.2	10.4	14.4	12.1	12.7	10.7	19.4	15	(°C)	

(1) Create a frequency distribution table where the class interval is 14°C to less than 16°C, with each class having a width of 2°C.
(2) Based on the frequency distribution table created in step (1), draw a histogram.

3 The following table summarizes the functions available in spreadsheet software. Choose the term that best fits into the blanks [1] to [3] from the options **A** to **D** below, and answer using the letters.

Function	Explanation
([1])	Determines the mean.
([2])	Determines the median.
([3])	Determines the mode.

A MODE **B** AVERAGE **C** MEDIAN **D** VARP

Exercise

1 Answer the following questions.

(1) The following data records the heights of 30 students in a class. Answer the following questions.

168	151	168	166	183	162	167	176	161	175	
156	165	173	152	165	177	156	171	158	172	
172	148	146	167	173	155	169	180	161	163	(cm)

[1] Create a frequency distribution table where the class interval is 145 cm to less than 150 cm, with each class having a width of 5 cm.

[2] Based on the frequency distribution table created in question [1], draw a histogram.

(2) The following data represents the daily sleep duration of 20 high school students. Answer the following questions.

7	7	5	9	7	6	8	4	8	10	
3	8	6	8	9	5	8	6	8	8	(Time)

[1] Determine the mean of the data.

[2] Determine the median of the data.

[3] Determine the mode of the data.

2 Items **A** to **C** below describe functions available in spreadsheet software. Mark "○" if the statement is appropriate, and "×" if it is inappropriate.

A The mean can be determined using the AVERAGE function in spreadsheet software.

B The median can be determined using the MEDIAN function in spreadsheet software.

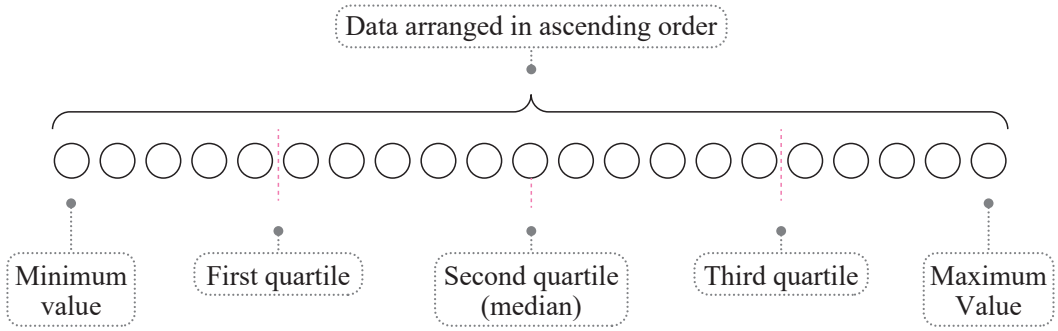
C The mode can be determined using the SUM function in spreadsheet software.

10-3 Data Analysis [2]

Point!

1 Quartiles

(1) ([1]**Quartile**) : The numbers that divide the data into four equal parts when arranged in ascending order. Quartiles are referred to as follows in ascending order: ([2]**first quartile**), ([3]**second quartile**), ([4]**third quartile**). The second quartile is the ([5]**median**).



(2) Steps for determining the quartiles

[1] Arrange the data in ascending order and determine the ([6]**median**).

[2] With the median in [1] as the boundary, divide into two groups: ([7]**less than or equal to the median**) and ([8]**greater than or equal to the median**).

[3] For each group created in [2], determine the ([9]**median**).

2 Box-and-whisker Plots

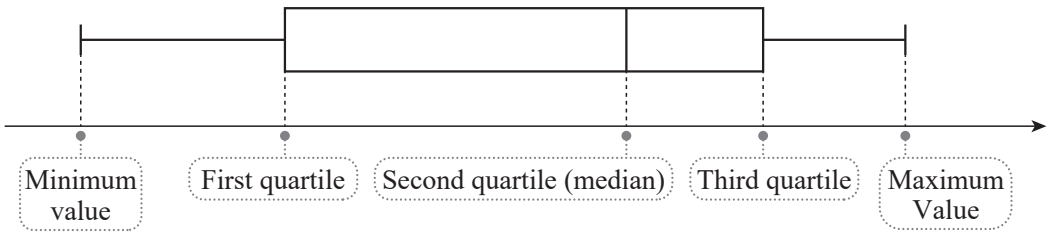
(1) ([10]**Box-and-whisker plot**) : A figure using a box and lines (whiskers) to represent the minimum value, first quartile, second quartile, third quartile, and maximum value in a single figure, as shown below.

(2) Steps to draw a box-and-whisker plot

[1] Determine the ([11]**minimum value**), ([12]**maximum value**), and ([13]**quartiles**), and draw five vertical lines representing the points.

[2] Draw a box with the first quartile as the left edge and the third quartile as the right edge.

[3] Draw a horizontal line from each end of the box to the minimum and maximum values.



*A box-and-whisker plot can also be drawn vertically.

Warm Up

Answer the following questions.

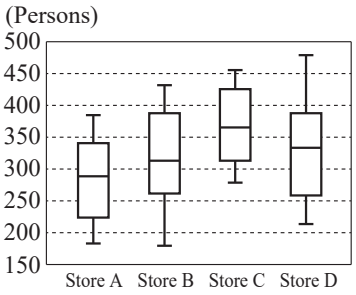
(1) The following data represents the math test scores of ten students.

58 72 52 61 36 43 20 32 38 47 (Points)

[1] Provide the first quartile, second quartile, and third quartile.

[2] Draw a box-and-whisker plot based on this data.

(2) The figure on the right shows data on the number of daily customers at stores A, B, C, and D over a span of 31 days in a box-and-whisker plot.



[1] List all stores with days where the number of customers exceeded 450.

[2] Provide the minimum number of days where the number of customers exceeded 300 at store B.

[3] List all stores that had eight or more days with fewer than 250 customers.

Explanation

(1) [1] When you arrange the data in ascending order,

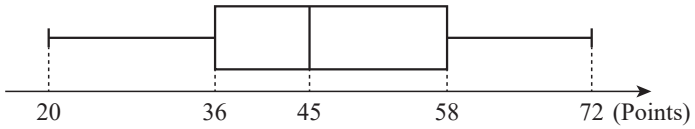
20, 32, 36, 38, 43, 47, 52, 58, 61, 72

The median is (43 + 47) ÷ 2 = 45 (points).

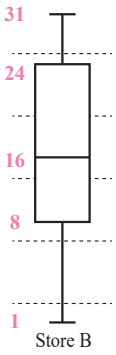
The median of the group less than or equal to the overall median, 20, 32, 36, 38, 43, is 36. The median of the group greater than or equal to the overall median, 47, 52, 58, 61, 72, is 58.

Therefore, the first quartile is 36 points, the second quartile is 45 points, and the third quartile is 58 points.

[2] The minimum value is 20, the maximum value is 72, the first quartile is 36, the second quartile is 45, and the third quartile is 58, so the box-and-whisker plot will look like the figure below.



(2) Since the data was collected over 31 days, the median is the 16th value, the first quartile is the 8th value, and the third quartile is the 24th value. Accordingly, write it into the figure as shown on the right.



[1] Choose the box-and-whisker plots where the maximum exceeds 450, which would be store C and store D.

[2] From the box-and-whisker plot of store B, it can be determined that the median exceeds 300 people. From the figure on the right, the median corresponds to the value of the 16th data point when counted from smallest to largest. Therefore, from the 16th to the 31st data points, the number exceeds 300 people, so it is 16 days.

[3] From the figure on the right, the first quartile is the 8th data point when counted from the smallest. Therefore, we should select the box-and-whisker plot where the first quartile is less than 250 people, so store A.

Try

1 The following data represents the test scores of ten individuals. Answer the following questions.

68 90 75 64 72 32 46 86 59 82 (Points)

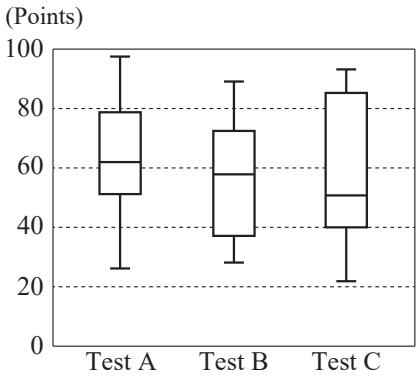
(1) Provide the first quartile, second quartile, and third quartile.

(2) Draw a box-and-whisker plot based on this data.

2 The figure on the right is a box-and-whisker plot representing the scores of tests A, B, and C taken by 50 first-year high school students. Answer the following questions.

(1) Identify all tests where there are 13 or more students scoring 80 or above.

(2) Identify all tests where more than half of the students scored below 60 points.



Exercise

1 The following data represents the Japanese language test scores of 10 individuals. Answer the following questions.

66 77 79 46 30 25 54 95 50 86 (Points)

(1) Provide the first quartile, second quartile, and third quartile.

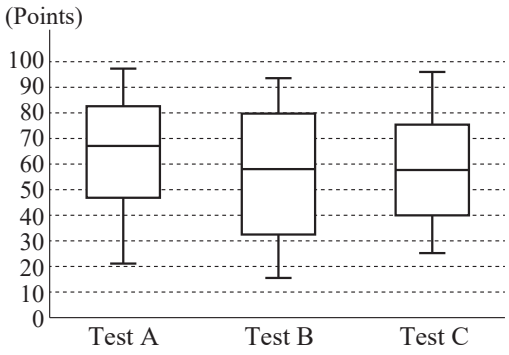
(2) Draw a box-and-whisker plot based on this data.

2 The figure on the right is a box-and-whisker plot representing the scores of tests A, B, and C taken by a class of 30 high school students.

Answer the following questions.

(1) Identify all tests where there are 8 or more students scoring 80 or above.

(2) Identify all tests where more than half of the students scored below 60 points.



10-4 Data Analysis [3]

Point!

Variance and Standard Deviation

• Variance and standard deviation: These represent numerically how much the data is spread out from the mean ([1]**mean**). Let x_1, x_2, …, x_n represent the values of the data. $\bar{x}$ refers to the mean, and n refers to the number of data points.

(1) ([2]**Deviation**) : $x_1 - \bar{x}$, $x_2 - \bar{x}$, …, $x_n - \bar{x}$

The deviation is the value obtained by subtracting the mean from a data value.

(2) ([3]**Variance**) : The mean value of $(x - \bar{x})^2$. The ([4]**larger**) the variance, the more dispersed the data is.

In Google Sheets, the VARP function can be used to determine variance, while in Excel, the VAR.P function is used.

(3) ([5]**Standard deviation**) : $s = \sqrt{\text{Variance}}$

The standard deviation is the positive square root of the variance.

In spreadsheet software, the STDEV.P function can be used to determine the variance.

The standard deviation should be expressed in the same unit as the data.

<Steps for Calculating the Standard Deviation>

Summarize data in a table as shown on the right according to the following steps.

[1] Determine the ([6]**mean**) of the data.

[2] Find the ([7]**deviation**) for each individual data point.

[3] ([8]**Square the deviation**) of each individual data point.

[4] Determine the variance, which is the ([9]**mean**) of [3].

	x	$x - \bar{x}$	$(x - \bar{x})^2$
x_1		[2]	[3]
x_2		[2]	[3]
x_3		[2]	[3]
Total			
Mean	[1]		[4]

Variance

Warm Up

The following data represents the results of a handball throw by six individuals.

26, 25, 32, 28, 32, 25 (m)

(1) Determine the mean $\bar{x}$ of the data.

(2) Determine the variance and standard deviation of the data.

Explanation

(1) Summarize the data in a table like the one on the right.
The mean is 28 m.

(2) The variance is 9, and the standard deviation is $\sqrt{9} = 3$ (m).
3 m

	x	$x - \bar{x}$	$(x - \bar{x})^2$
x_1	26	-2	4
x_2	25	-3	9
x_3	32	4	16
x_4	28	0	0
x_5	32	4	16
x_6	25	-3	9
Total	168		54
Mean	28		9

Try

1 The following data represents the English vocabulary test scores of ten students.

9, 3, 4, 10, 10, 5, 7, 9, 10, 3 (points)

(1) Determine the mean of the data.

(2) Determine the variance and standard deviation of the data to two decimal places. However, let $\sqrt{2}$ = 1.414.

2 The following table shows the number of days it took five students to complete two types of assignments, A and B.

A	22	28	25	26	24
B	21	29	27	25	28

(1) Determine the variance for assignment A.

(2) Determine the standard deviation to two decimal places for assignment B. However, let $\sqrt{2}$ = 1.414.

(3) Answer which data set, assignment A or assignment B, has a greater variance.

Exercise

1 The following data represents the weights of eight harvested strawberries.

22, 25, 18, 17, 22, 21, 20, 15 (g)

(1) Determine the mean of the data.

(2) Determine the variance and standard deviation of the data.

2 The following table shows the test scores in math and English for five students.

Math	7	9	6	10	8
English	10	8	6	4	2

(1) Determine the variance of the math scores.

(2) Determine the standard deviation of the English test scores to two decimal places. However, let $\sqrt{2}$ = 1.414.

(3) Answer which data set has a greater variance, math or English.

10-5 Data Analysis [4]

Point!

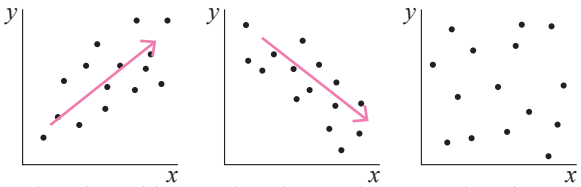
Scatter Plots, Correlations, and Cross Tabulation

(1) ([1]**Scatter plot**) : A graph that represents points on a plane using pairs of variables, x and y, as coordinates.

[1] When there is a tendency for both corresponding x and y to increase with each other, it is referred to as a ([2]**positive correlation**).

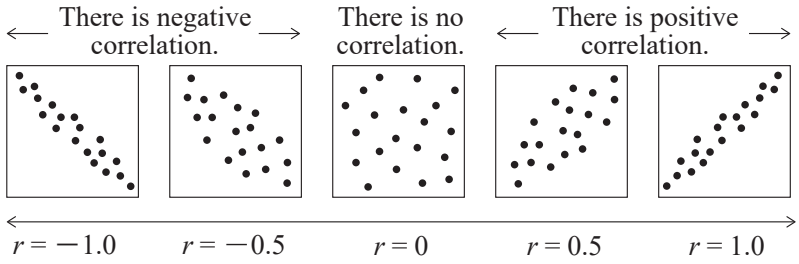
[2] When there is a tendency for one of the corresponding x and y to decrease as the other increases, it is referred to as a ([3]**negative correlation**).

[3] When there is neither a positive nor a negative correlation between the corresponding x and y, it is referred to as ([4]**no correlation**).



(2) ([5]**Correlation coefficient**) : An indicator to measure the strength of a correlation.

- The correlation coefficient r is represented by ([6] $-1 \leq r \leq 1$).
- The correlation coefficient r indicates a stronger ([7]**positive correlation**) as it approaches 1, and a stronger ([8]**negative correlation**) as it approaches -1.



- To determine the correlation coefficient using spreadsheet software, you can use the CORREL function.

(3) ([9]**Causal relationship**) : A relationship where one of two things is the cause and the other is the effect.
<Example> When the temperature rises, sales of shaved ice increase.

(4) ([10]**Pseudo-correlation**) : When two things appear to have a causal relationship, even though there is none.
<Example> There is no causal relationship between "As the temperature rises, shaved ice sales increase" and "As the temperature rises, the number of heatstroke cases increases."

(5) ([11]**Cross tabulation**) : An aggregation method for comparing data among two or more categories.

<Example>

	Agree	Oppose	Neither
Male	57	97	46
Female	73	75	52
Total	130	172	98

Warm Up

Answer the following questions.

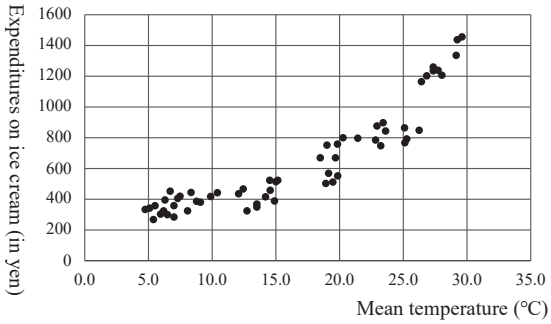
(1) There is data consisting of the following two variables, x and y.

x	7	4	6	2	9	3	8	1	6	3
y	3	6	5	8	3	4	2	9	2	5

[1] Create a scatter plot.

[2] Describe the tendency observed between x and y.

(2) The relationship between mean temperature and ice cream expenditures was investigated and summarized in the scatter plot on the right.



[1] Choose the description representing the tendency of the two from the options **A** to **D** below, and answer using the letter.

A There is a weak negative correlation.

B There is a strong negative correlation.

C There is a weak positive correlation.

D There is a strong positive correlation.

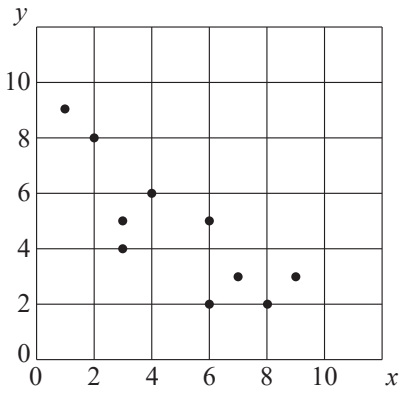
[2] Choose the number that best fits into the blank in the following sentence from the options **A** to **D** below, and answer using the letter.

The correlation coefficient between the mean temperature and ice cream expenditures was ( ).

A -0.9 **B** -0.4 **C** 0.4 **D** 0.9

Explanation

(1) [1] The scatter plot is shown below.



[2] There is a negative correlation according to the scatter plot.

(2) [1] Based on the scatter plot, it can be interpreted as having a “strong positive correlation” because it forms a linear upward graph. Therefore, **D**.

[2] Based on [1], since it is known that there is a strong positive correlation between mean temperature and ice cream expenditure, it can be inferred that the highest value of 0.9 is most appropriate. Therefore, **D**.

Try

1 There is data consisting of the following two variables, x and y.

x	7	4	6	2	8	3	8	1	6	3
y	3	6	5	8	3	4	2	8	2	5

(1) Create a scatter plot.

(2) Describe the tendency observed between x and y.

2 To examine the relationship between study time, commuting time, and test scores, a survey was taken by 40 students in a class regarding their daily study time on weekdays. Below are scatter plots representing the relationship between each student's study time and test scores (Figure 1), and the relationship between commuting time and test scores (Figure 2). Answer the following questions.

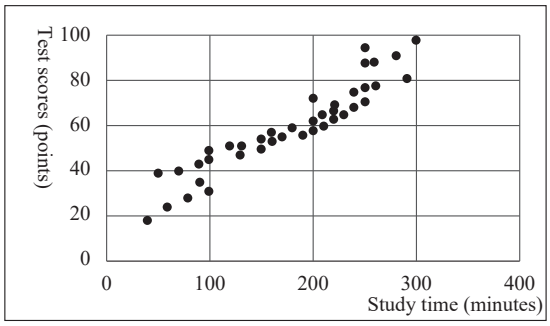


Figure 1: Scatter plot of study time and test scores

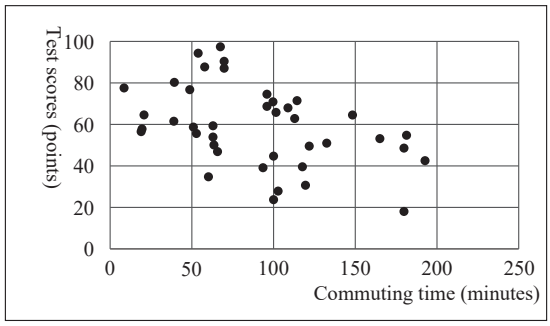


Figure 2: Scatter plot of commuting time and test scores

(1) Compare the two scatter plots, and choose the trend observed in each from the options **A** to **D** below, and answer using the letter.

A Figure 1 shows a strong positive correlation, while Figure 2 shows a strong negative correlation.

B Figure 1 shows a strong positive correlation, while Figure 2 shows a weak negative correlation.

C Figure 1 shows a weak positive correlation, while Figure 2 exhibits a strong negative correlation.

D Figure 1 shows a weak positive correlation, while Figure 2 shows a weak negative correlation.

(2) Choose the number that best fits into the blanks [1] and [2] in the following sentence from the options **A** to **D** below, and answer using the letters.

> When the correlation coefficients for study time, commuting time, and test scores are determined, the correlation coefficient between study time and test scores is ([1]), and the correlation coefficient between commuting time and test scores is ([2]).

A -0.9 **B** -0.4 **C** 0.4 **D** 0.9

3 The table on the right shows the results of a survey taken by 50 students in a class regarding whether they like Japanese, math, and English. Fill in the appropriate values for blanks [1] to [7] in the table.

	Like	Dislike	Neither	Total
Japanese	2	([1])	([2])	([3])
Math	6	([4])	5	13
English	([5])	([6])	1	18
Total	15	21	([7])	50

Exercise

1 There is data consisting of the following two variables, x and y.

x	6	4	7	5	8	6	2	9	8	5
y	7	10	2	4	3	4	4	5	4	7

(1) Create a scatter plot.

(2) Describe the tendency observed between x and y.

2 The rent of rental properties around a station was investigated to examine the relationship with the size of the room and the distance to the nearest station. The following are scatter plots representing the relationship between the size of each room and the rent (Figure 1), and the relationship between the distance to the nearest station from each room and the rent (Figure 2). Answer the following questions.

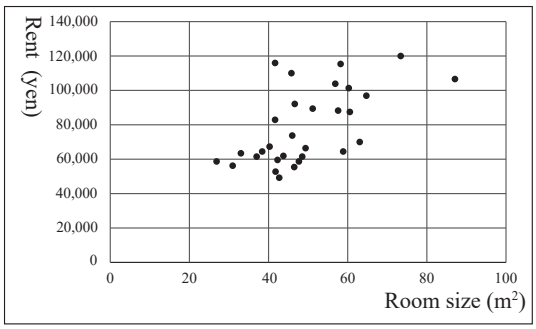


Figure 1: Scatter plot of room size and rent

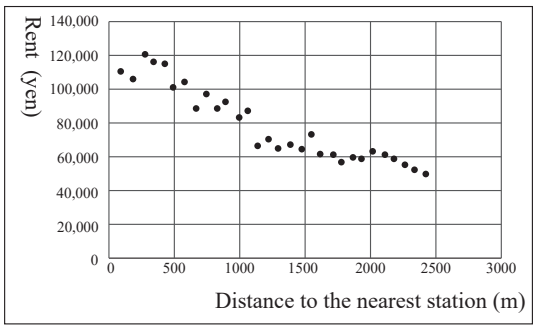


Figure 2: Scatter plot of rent and distance to nearest station

(1) Compare the two scatter plots and provide the tendencies that can be observed in each. Choose one from the options **A** to **D** below, and answer using the letter.

A Figure 1 shows a strong positive correlation, while Figure 2 shows a strong negative correlation.

B Figure 1 shows a weak positive correlation, while Figure 2 exhibits a strong negative correlation.

C Figure 1 shows a strong positive correlation, while Figure 2 shows a weak negative correlation.

D Figure 1 shows a weak positive correlation, while Figure 2 shows a weak negative correlation.

(2) Choose the number that best fits into the blanks [1] and [2] in the following sentence from the options **A** to **D** below, and answer using the letters.

> When the correlation coefficients for room size and rent, and distance to the nearest station and rent are determined, the correlation coefficient between room size and rent is ([1]), and the correlation coefficient between the distance to the nearest station and rent is ([2]).

A -0.9 **B** -0.6 **C** 0.6 **D** 0.9

3 The table on the right shows the results of a survey taken by 200 school students regarding their main means of commuting to and from school, categorized by grade. Fill in the appropriate values for blanks [1] to [7] in the table.

	On foot	Bicycle	Other	Total
Grade 1	29	33	([1])	([2])
Grade 2	([3])	([4])	([5])	67
Grade 3	([6])	32	9	66
Total	([7])	94	24	200

10-6 Data Analysis [5]

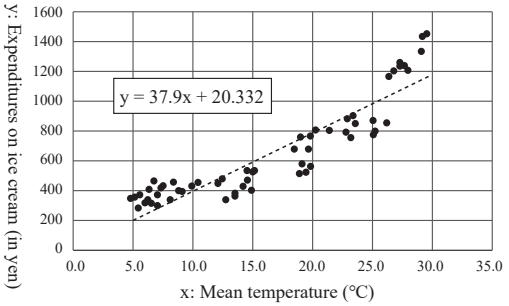
Point!

Regression Analysis

(1) ([1]**Regression analysis**) : A method to investigate and clarify the relationship between a resulting value and a causal value. It is used in predictions and simulations.

(2) ([2]**Simple regression analysis**) : Predicts the outcome variable y using the causal variable x.

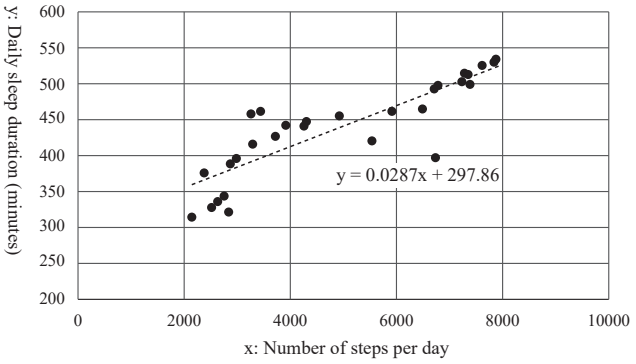
(3) ([3]**Regression line**) : Refers to the line used to determine predicted values in a scatter plot. A regression line can be expressed with the ([4]**regression line equation**) in the form of a first-degree function y = ax + b.



(4) ([5]**Least squares method**) : A method used to ([6]**minimize**) the vertical error, or the ([7]**residual**), between the actual data points and the regression line.

Warm Up

The following scatter plot summarizes the relationship between the number of steps and the sleep duration for a student over the course of one month. Answer the following questions about this graph.



(1) What is the term for describing the process of determining the equation for a straight line in the form y = ax + b representing the relationship between two variables, explanatory variable x and resulting variable y?

(2) Complete the following sentences by filling in the blanks [1] to [3] with the appropriate items.

> When representing the relationship between two variables with a straight line, the ([1]) method is for finding the line that best fits the actual data. In the ([1]) method, a straight line is determined to ([2]) the error, referred to as the ([3]), between each data point and the line.

(3) The straight line in the figure represents the relationship between two variables using a straight line. If the equation of this line is "y = 0.0287x + 297.86," what is the estimated sleep duration in minutes when the number of steps per day is 5,000? Give the answer as an integer (rounded off to the nearest whole number).

Explanation

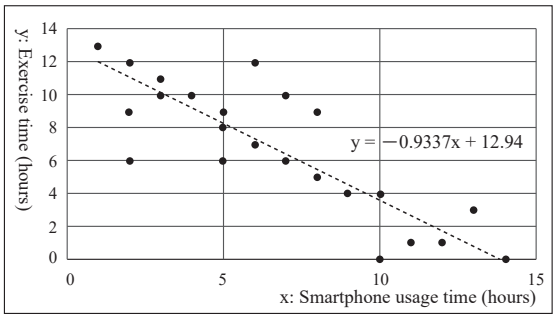
(1) Simple regression analysis

(2) [1] least squares [2] residual [3] minimize

(3) Substituting x = 5,000 into y = 0.0287x + 297.86, we get
y = 0.0287 × 5,000 + 297.86 = 441.36 ≈ 441
Therefore, the sleep duration is estimated at 441 minutes.

Try

The following figure is a scatter plot summarizing the results of a survey taken by 30 students from a class regarding their smartphone usage time and exercise time per day. Refer to this graph and answer the following questions.



(1) What is the term for the equation for a straight line in the form y = ax + b representing the relationship between two variables: causal variable x and resulting variable y?

(2) The straight line in the figure represents the relationship between two variables using a straight line. If the equation for this line is "y = − 0.9337x + 12.94," what is the estimated exercise time when smartphone usage is 10 hours? Give the answer as an integer (rounded off to the nearest whole number).

(3) Choose the term or number that best fits into the blanks [1] and [2] in the following sentences from the options **A** to **D** below.

> The least squares method is used to minimize the error between a straight line representing the relationship between two variables and the actual data. This type of error is referred to as ([1]). If the absolute value of the correlation coefficient is ([2]), the value obtained from the straight line determined by the least squares method is considered an estimate.

A Residual **B** Deviation **C** 1 **D** Other than 1

Exercise

1 The following figure is a scatter plot summarizing the measurement results of the height and weight of 100 adults. Refer to this graph and answer the following questions.

(1) Choose the one that best fits into the blanks [1] and [2] in the following sentences from the options **A** to **D** below.

> Based on the scatter plot, there is a tendency for weight to increase as height increases. Accordingly, when ([1]), it is referred to as a ([2]) between the two variables.

A Positive correlation
B Negative correlation
C One value increases and the other value also increases
D One value increases and the other value decreases

(2) The straight line in the figure represents the relationship between two variables using a straight line. If the equation for the line is "y = 0.5511x − 32.958," what is the approximate weight when the height is 173 cm? Give the answer as an integer (rounded off to the nearest whole number).